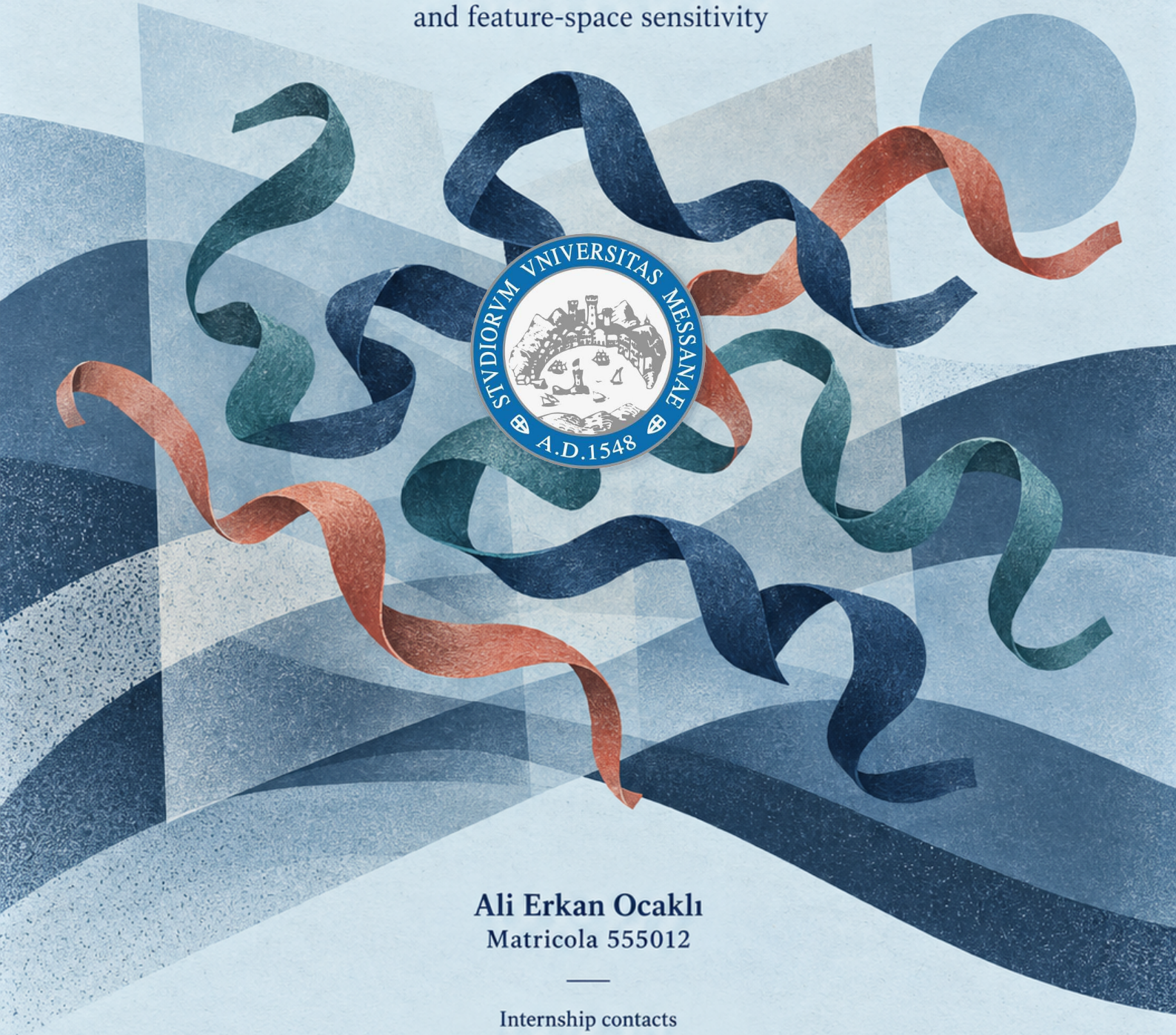
UNIVERSITY OF MESSINA
Data Analysis L-31

CURRICULAR INTERNSHIP REPORT

# RobustCluster-Aβ

## Reproducible analysis of amyloid-β40 conformational descriptors

Scientific code audit, sampling diagnostics and feature-space sensitivity




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# Abstract

Radius of gyration and solvent-accessible surface area (SASA) describe related aspects of peptide shape. Their correlation, however, does not determine whether one can be omitted from a clustering model. This internship report investigates that distinction in amyloid-β40 descriptors and audits the analysis supplied for reproducing the workflow of Cadenelli and colleagues.

The dataset contains 300,003 observations in three replica segments, covering the source-labelled 1–3 microsecond interval. Inspection of the original script identified problems with parsing, residual clusters, free-energy interpolation, replica pooling and weighted summaries. These were addressed in a modular Python implementation with numerical tests and recorded configurations. A recovered free-energy surface passed grid-completeness and query checks. Its energy units and construction provenance remain unresolved, so the reported experiments use the biased descriptors without equilibrium reweighting.

The pooled Rg–SASA correlation is 0.848. The first two standardized principal components explain 88.1% of feature variance. To test the consequence of removing SASA, an independent experiment fitted Gaussian mixtures to four descriptors and to the remaining three. A shared replica-balanced fitting sample was used for five component counts and five initialization seeds, giving fifty fits. All reached the numerical convergence criterion. With three components, the median cross-representation Adjusted Rand index was 0.864; it fell to 0.457 with six components. Agreement therefore depended on the chosen model resolution, despite the strong global feature correlation.

Temporal diagnostics place a separate limit on interpretation: every replica–descriptor series failed at least one correlation-time reliability criterion. The Gaussian-mixture comparison consequently remains a descriptive analysis of one fixed fitting sample, evaluated on observations that include the fitting rows. It does not identify physical metastable states or unbiased kinetic rates.

The resulting software and report make the completed descriptor analysis inspectable and provide a tested basis for further reproduction. Verified FES provenance is still required before equilibrium conformational populations can be reported.

**Keywords:** amyloid-β40; intrinsically disordered peptides; well-tempered metadynamics; density peaks; statistical reweighting; reproducibility.

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# Abbreviations and evidence categories

| Term | Meaning |
|---|---|
| ACF | Autocorrelation function |
| ARI | Adjusted Rand index |
| CV | Collective variable |
| GMM | Gaussian mixture model |
| DPC / DPA | Density Peak Clustering / Density Peaks Advanced |
| ESS | Effective sample size; estimator and assumptions stated explicitly |
| FES | Free-energy surface |
| IDP | Intrinsically disordered protein or peptide |
| MD | Molecular dynamics |
| PCA | Principal component analysis |
| SASA | Solvent-accessible surface area |
| SEM | Standard error of the mean |
| WT-MetaD | Well-tempered metadynamics |

Published methodology, the implemented reproduction, corrections to supplied code, robustness extensions, and future work are distinct evidence categories throughout this report. A successful software test demonstrates behavior on a specified fixture; it does not establish a biological result. A BLOCKED analysis lacks required inputs or provenance and is not presented as completed. The report distinguishes assigned learning topics, demonstrable project outputs, and analyses that remain unperformed.

# 1 Introduction and internship objectives

## 1.1 Biological and computational motivation

How should a collection of peptide configurations be reduced to a few interpretable groups? For amyloid-β monomers, the choice is consequential: configurations of similar size can differ in secondary structure, solvent exposure and contacts. Intrinsic disorder describes an ensemble with structural organization, even when no single reference conformation is sufficient [1]. The analysis must therefore connect a compact numerical representation to the molecular properties it is meant to summarize.

Cadenelli and colleagues address this problem by comparing monomeric Aβ40 and Aβ42 ensembles under a common simulation and analysis protocol [2]. The material supplied for this internship contains Aβ40 descriptors only. It supports an examination of the analysis choices, rather than a new comparison between isoforms. Related energy-landscape studies provide the biological motivation for characterizing monomers, while aggregation itself remains a distinct process from the geometric clustering studied here [3].

## 1.2 Internship context and objectives

The project forms part of the Data Analysis L-31 internship at the University of Messina, with Andrea Ciccolo and Valeria Conti Nibali as internship contacts. The documented programme moved from protein structure and intrinsic disorder to force fields, molecular dynamics and clustering. Its final-stage brief called for an independent method or analysis drawing on the degree programme.

The first task was to audit the supplied script and inputs so that the published analysis could be implemented with documented corrections. A second question emerged from the descriptor analysis: radius of gyration and SASA are correlated, but does removing SASA preserve the fitted groups? The independent experiment tests that question across Gaussian-mixture component counts and initializations. It can be performed on the supplied descriptors without first resolving the equilibrium reweighting.

## 1.3 Contributions and evidence boundaries

The completed work comprises a source audit, tested numerical routines, descriptor and sampling diagnostics, FES grid validation, and a Gaussian-mixture feature-ablation pilot. The pilot is the original analysis developed for this report; the modular clustering implementation supports a separate reproduction of the published method. That reproduction remains incomplete because the FES energy units and construction provenance have not been established.

Each calculation records its inputs, settings, software versions and outputs. These records make it possible to trace a scientific plot to the table that generated it and to distinguish a successful run from an unavailable analysis. The arrangement follows research-software guidance on portable workflows and machine-actionable provenance [4–6]. It also gives the report a practical test of reproducibility: another reader should be able to inspect how each reported observation was obtained.

# 2 Scientific and theoretical background

## 2.1 Amyloid-β40 and intrinsically disordered ensembles

An ensemble description retains both conformations and their statistical weights. For an IDP, a representative structure shows one member of that ensemble; an observable measured over the ensemble combines many such members. Global descriptors make this information manageable, at the cost of detail. Two configurations can have the same radius of gyration and solvent-accessible area but different residue contacts. A cluster built from those descriptors identifies similarity under that representation, rather than a unique atomic fold.

Experimental constraints can improve the characterization of such ensembles, but observables such as NMR or SAXS measurements generally average over conformations. Borthakur and colleagues describe a maximum-entropy procedure that combines simulations with experimental data to constrain IDP ensembles [7]. That study suggests a possible external comparison. Such a comparison requires compatible structural data, observable definitions and statistical weights.

## 2.2 Molecular dynamics and conformational ensembles

Molecular dynamics generates configurations by integrating the motion of a model system under a specified potential-energy function. Force-field parameters, solvent representation, thermodynamic conditions and sampling all affect the resulting ensemble. The primary study used GROMACS, CHARMM36m and PLUMED [2, 8–10]. The work reported here begins downstream of those simulations, with the supplied descriptor table.

In an equilibrium canonical ensemble, an observable $A(q)$ has expectation proportional to the integral of $A(q)\exp[-\beta_T U(q)]$, where $q$ denotes coordinates and $U$ is potential energy. For energies per molecule, $\beta_T = 1/(k_B T)$; for the molar energies used in the implementation, the consistent factor is $1/(RT)$. Mixing these energy conventions introduces a scale error even when the reweighting code is numerically stable.

## 2.3 Sampling limitations

Saving configurations more frequently increases the number of rows in a trajectory without necessarily increasing the amount of independent information. Nearby frames can remain in the same slowly changing region of descriptor space. Under suitable assumptions, statistical inefficiency describes the resulting inflation of estimator variance. It cannot compensate for a region of the ensemble that was never explored. Even a stable overall mean can conceal changes in the occupation of different structural subpopulations.

Assessing equilibration and sampling quality requires evidence beyond a raw frame count [11, 12]. Here, an additional limitation is that the trajectory was generated under a changing bias. Correlation statistics are calculated as diagnostics of that recorded process. They are not identified with unbiased relaxation times, transition rates, or kinetic states. The fixed published analysis interval is retained for reproduction; the extension assesses its statistical behavior rather than silently choosing a new interval.

## 2.4 Well-tempered metadynamics

WT-MetaD adds a history-dependent potential along selected CVs. Its tempering parameter is conventionally expressed as $\gamma = (T + \Delta T)/T > 1$. In the long-time regime, the bias approaches a scaled negative free energy, up to an additive constant [13]. This relationship motivates the final-bias approximation used in the primary paper.

The same bias that promotes exploration also changes the sampled occupancies and the forces acting on the peptide. Population estimates therefore require a statistical correction, and the recorded time cannot

be read directly as an unbiased physical clock. Thermodynamic reweighting addresses the distribution under its stated assumptions; recovering kinetic information is a separate problem.

## 2.5 Gaussian hills and bias construction

For CV vector $s$, hills deposited at times $t_k$ generate a bias of the form

$$V(s,t) = \sum_{t_k<t} h_k \exp\left[-\frac{1}{2}\sum_a \frac{(s_a - s_a(t_k))^2}{\sigma_a^2}\right]. \tag{2.1}$$

In WT-MetaD, deposited heights decrease with the accumulated bias as $h_k = h_0 \exp[-V(s(t_k), t_k)/(k_B \Delta T)]$, using molecular energy units. For molar units, $k_B$ is replaced by $R$. The precise convention must agree with the simulation software and energy units. Equation 2.1 describes the methodological construction, not a reconstruction performed in this project. No project HILLS file is available.

PLUMED distinguishes the instantaneous bias output, the time-dependent offset, and their difference. Its `metad.rbias` component equals `metad.bias` minus `metad.rct`; the latter represents $c(t)$ [14]. A column named `metad.rbias` is therefore potentially useful for a separate time-dependent reweighting analysis. Its name alone does not establish that the offset was computed with the required settings or that its units are known.

## 2.6 Free-energy surfaces

For a CV distribution $p(s)$, free energy is $F(s) = -\beta_T^{-1} \log p(s) + C$, where $C$ sets an arbitrary reference. Free energy is a property of the chosen projection; integrating out other coordinates can obscure structural distinctions. A two-dimensional minimum does not uniquely specify a three-dimensional conformation.

PLUMED's `sum_hills` supports reconstruction from accumulated hill records and distinguishes free-energy output from negative-bias output [15]. That distinction matters because the WT scaling factor must not be applied twice. The recovered complete file labels its energy field `file.free`, but the energy unit, generating commands, and replica-combination procedure are still needed for a verified final-bias calculation.

## 2.7 Statistical reweighting

For a quasi-static bias, the correction from a biased distribution to the corresponding unbiased distribution has weights proportional to $\exp(\beta_T V)$. Combining this with the WT relationship yields

$$\ell_i \equiv \log w_i = -\frac{1 - 1/\gamma}{RT} F(\alpha_i, \beta_i). \tag{2.2}$$

The free-energy zero does not affect the normalized result: adding a constant to $F$ adds the same offset to all relevant log weights, which cancels on normalization. Equation 2.2 implements the primary paper's final-bias approximation. The central assumption is that the analyzed interval admits the quasi-stationary interpretation needed for this correction.

Alternative MetaD reweighting approaches account explicitly for changing bias normalization [16, 17]. For verified PLUMED offset-corrected output, the optional comparison uses $\ell_i = \mathrm{rbias}_i/(RT)$ [18]. These two estimators are not interchangeable labels for the same computation. The current repository keeps their configurations and resulting outputs separate.

## 2.8 Structural descriptors

The baseline uses $n_\alpha$, $n_\beta$, $R_g$, and total SASA. In the primary study, the first two are scaled continuous secondary-structure CVs, intended to approximate residue counts. They should not be interpreted as exact integer DSSP assignments. DSSP itself classifies local structural patterns using geometrical and

hydrogen-bond information [19]; reproducing such assignments requires coordinates, which are absent here.

Radius of gyration summarizes global compaction; SASA measures solvent exposure. A correlation between them is plausible, but it leaves open whether each contributes useful distinctions within a cluster. Apolar and polar SASA and intra- and peptide–solvent hydrogen bonds provide further descriptors for characterization after baseline clustering. Their exact upstream definitions remain a provenance requirement, including whether the two SASA components are additive.

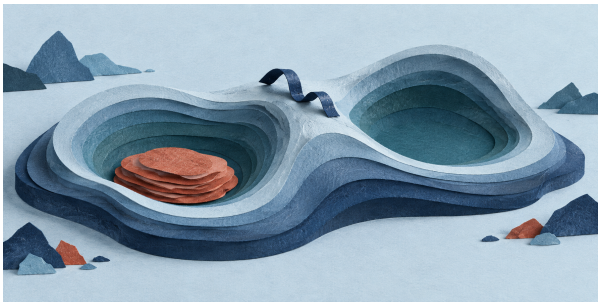
## 2.9 Density-based conformational clustering

Density-peak methods seek centers that combine high local density with separation from denser points. The primary study adapts that logic to recursively discover acceptable clusters across random subsets and density scales [2]. A silhouette coefficient summarizes the relation between within-cluster cohesion and separation from other clusters [20]. It is a geometric criterion, not proof of thermodynamic metastability.

Consensus then groups local clusters by similarities in their descriptor distributions. DPA offers a distinct, statistically motivated density-topography approach [21]. A DPA benchmark would test whether an alternative treatment of density produces similar families. Likewise, weighted ShapeGMM addresses structural ensembles in a coordinate-based representation [22]; its applicability here depends on obtaining valid structural frames.



**(a)** Conformational diversity



**(b)** An evolving sampling landscape

**Figure 1:** Conceptual editorial illustrations. Open paper ribbons evoke different conformations of a flexible peptide; layered deposits evoke the accumulation of metadynamics bias. Neither image depicts molecular coordinates, secondary-structure assignments, a measured FES or a physical trajectory. The qualitative ideas are formalized in the surrounding text and eq. (2.1). Generated-art provenance is documented in Appendix D.

# 3 Data and computational methodology

## 3.1 Data provenance

The input is a 10-column text table with 300 003 numerical rows. Its SHA256 and the hash of the original script are recorded in the inventory. The original materials are retained unchanged. The table contains scalar descriptors and a column labelled `metad.rbias`. Contact maps or molecular renderings would additionally require atomic coordinates and a verified mapping from each row to its structural frame.

The paper and the script serve different purposes in the audit. The publisher PDF describes the method to reproduce; the supplied source shows how that method was implemented. Execution evidence is needed to determine whether an analysis actually completed. Scientific V2 and its changelog were unavailable, so their contents remain an explicit gap in the method map.

## 3.2 Replica segmentation

Replica boundaries are detected where the first column decreases. Parsing requires exactly three segments, strictly increasing times within each segment, finite numerical entries, and a recognized header. Extra segments, duplicate times, malformed rows, or missing interval coverage fail validation rather than being silently skipped.

Frame identities are preserved before any selection or shuffling: zero-based source row, one-based file line, replica ID and zero-based frame index within its original segment. A partition comparison then matches the same observations across runs. Comparing arrays in their current storage order would otherwise confuse a reordering of frames with a change in the clustering.

## 3.3 Analysis interval

The baseline interval includes both endpoints from 1 000 000 to 3 000 000 in the source-labelled picosecond time unit. Each replica contributes 100 001 frames separated by 20 ps. The original script and published protocol support this unit interpretation, but the upstream simulation/analysis metadata are absent. The interval therefore corresponds to the supplied 1–3 µs window under a recorded provenance assumption.

The published workflow excludes the first microsecond. The supplied table already begins at the selected boundary, so it permits verification of interval coverage but cannot support an independent assessment of the preceding equilibration period or the full-trajectory convergence analysis.

## 3.4 Descriptor definitions

The canonical clustering columns are $n_\alpha$, $n_\beta$, $R_g$ in nm, and SASA in nm$^2$, with units inherited from the original script. Header spelling differences are mapped explicitly, including the original `HBONNDSPS` spelling. The `metad.rbias` column is preserved as a separate field and is not renamed to an interpolated free energy.

The FES and the clustering algorithm use different coordinates. FES queries apply the paper's descriptor conversions, $\alpha = n_\alpha/1.1$ and $\beta = n_\beta/2.5$, whereas clustering uses normalized features. Passing min–max-scaled values to the FES would select different physical CV locations. Separate modules and saved parameters keep these transformations distinct.

## 3.5 Feature normalization

For each baseline descriptor $x_f$, pooled analysis-window min–max scaling gives $z_f = (x_f - a_f)/(b_f - a_f)$, where $a_f$ and $b_f$ are the pooled minimum and maximum. Both values are saved. A constant feature is

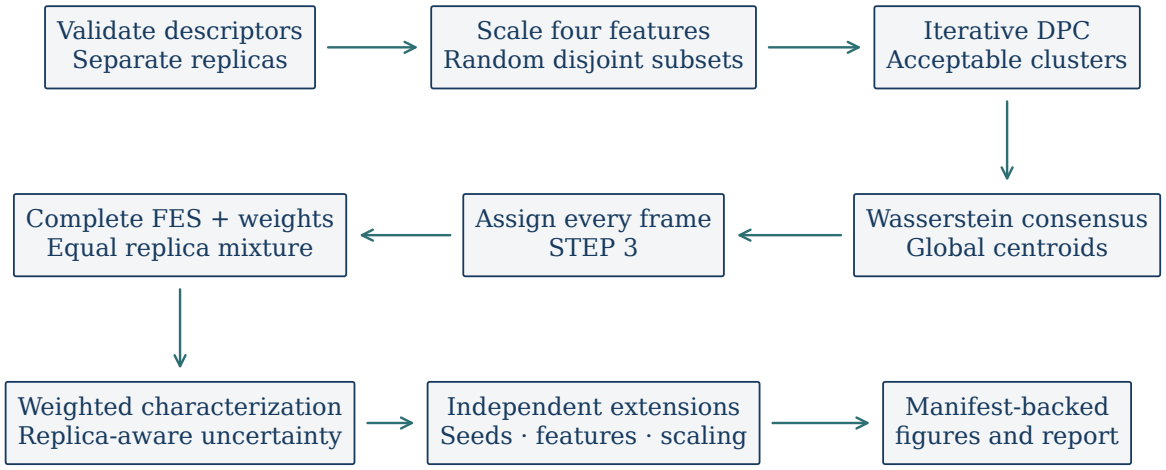



**Figure 2:** Implemented analysis design. The primary branch separates geometric cluster discovery from statistical reweighting. Robustness experiments rerun designated stages independently. The FES grid has been validated, while empirical weighting remains blocked by energy-unit and construction provenance; this workflow diagram documents implementation, not successful completion of every box.

an invalid input for this transform, rather than a reason to divide by zero or silently remove a scientific variable.

The pooled transform gives every replica and discovery group the same coordinate system. If each group were scaled independently, an observation's numerical position would depend on the group to which it had been randomly assigned. The baseline therefore retains the source's common min–max scaling. Standardization belongs to the separate GMM protocol; the planned DPC scaling comparisons remain unperformed.

## 3.6 STEP 1: iterative Density Peak Clustering

The workflow in fig. 2 begins with random disjoint subsets. Production group size is approximately 30 000. A tiny final subset of at most ten frames is redistributed over preceding groups, preserving every identity. Local RNG streams are derived from a recorded master seed. This removes the legacy fixed-seed-list limit, but changes exact random draws relative to the original global-RNG implementation [23].

Local density is the number of neighbors within $d_c$, including the point itself as in the supplied code. To handle equal densities reproducibly, points are ordered by density and then by local frame position. The separation $\delta_i$ is measured to the closest earlier point in that order; the first point uses the maximum pairwise distance. Ranking $\rho_i\delta_i$ then identifies candidate centers that combine density with separation. The tie rule matters because an earlier point need not be strictly denser.

For each cutoff, candidate center counts are scanned from 10 down to 2. The selected candidate maximizes the number of acceptable clusters; ties retain the earlier, higher-center-count candidate. Acceptance requires at least 300 members and mean silhouette at least 0.3. After accepted members are removed, $d_c$ is multiplied by 0.9. The process stops when fewer than 300 points remain or five consecutive iterations accept no cluster. Residual points remain unassigned at this stage.

Distances are computed in bounded blocks into a float32 matrix. Cutoff density is evaluated row by row without an additional floating kernel matrix. Silhouettes reuse that distance matrix, subject to the validity condition that the number of labels lies between two and $N - 1$ [24]. The algorithm still has quadratic storage and potentially substantial repeated silhouette cost; the changes reduce avoidable allocations rather than changing its asymptotic scaling.

## 3.7 STEP 2: Wasserstein consensus

Only acceptable local clusters enter consensus. For clusters $A$ and $B$, the implemented source metric is

$$D(A,B) = \left[\sum_{f=1}^{4} W_1(A_f, B_f)^2\right]^{1/2}. \tag{3.1}$$

Here $W_1$ is the one-dimensional first Wasserstein distance between empirical marginal descriptor distributions [25]. Equation 3.1 does not describe joint four-dimensional optimal transport. It can miss changes in dependence structure that leave every marginal distribution unchanged.

The reproduction configuration uses raw descriptors, matching the supplied code. This mixes descriptor scales and units in the aggregate norm; the paper does not fully resolve that scaling choice. The normalized alternative is retained as an explicit extension. Average-linkage agglomeration is followed by the source's cubic-spline elbow heuristic: smoothing parameter 5, with the maximum second derivative sought in the first 90% of merge steps. Fewer than five local clusters or an entirely flat merge-height curve yields an explicit failure instead of an invented cut.

## 3.8 STEP 3: final frame assignment

Each global centroid is the mean normalized feature vector of all accepted local-cluster members assigned to that family. All analysis-window frames, including residuals and frames outside reduced discovery groups, are assigned once to the nearest centroid. Distance ties use deterministic label ordering. Centroids are not iteratively refitted after reassignment: doing so would introduce a k-means refinement absent from the documented baseline.

The final labels are stored with the original frame identities. This lets descriptor statistics, stability comparisons and any later structural mapping refer to exactly the same observations. Their molecular interpretation remains limited by the chosen descriptor representation.

## 3.9 FES validation and interpolation

The FES parser requires explicit axis names, extrema, grid counts, and nonperiodic status. It verifies row count, finite values including supplied derivative columns, coordinate spacing, unique Cartesian pairs, and agreement with the header. Row order is not assumed. A partial-download extension is rejected before parsing.

Interpolation uses SciPy's linear rectilinear-grid interpolator with bounds checking enabled [26, 27]. Every query must fall inside both axes. A bilinear analytic surface containing a mixed $\alpha\beta$ term provides an exact test, including boundary queries and shuffled rows. Missing cells and out-of-bounds queries are errors; no extrapolation or clipping is permitted.

## 3.10 Statistical weights

Equation 2.2 is evaluated in log space. Within replica $r$, normalized weights are

$$\widetilde{w}_{ri} = \exp\left[\ell_{ri} - \operatorname{logsumexp}_j(\ell_{rj})\right], \qquad W_{ri} = \frac{\widetilde{w}_{ri}}{3}. \tag{3.2}$$

Equation 3.2 gives each replica equal total mass, independently of the arbitrary log-weight offset or differences in raw weight sums. The recorded raw exponential is diagnostic only. If it is not representable, the output includes an explicit overflow/underflow status; normalized calculations remain based on log weights.

The intended frame table contains replica, source row, file line, frame index, time, physical CVs, interpolated free energy, raw log weight, raw weight/status, within-replica weight, pooled weight, and final label. No such real weighted table is reported in this release because FES energy-unit and construction provenance remain unverified.

## 3.11 Weighted cluster characterization

For a cluster, summing the pooled weights gives its relevance. Restricting those weights to its members gives the conditional descriptor mean and width. The width is the square root of the weighted central second moment: it describes heterogeneity within the family. Replica SEM addresses a different question, the variation of the relevance estimate across replicas, and is calculated from the separate replica populations.

The uncertainty extension keeps the fitted labels and supplied FES fixed while resampling contiguous observation blocks independently within each replica. It renormalizes the selected weights and recalculates populations for each draw. The resulting intervals would be conditional on that partition and FES; they do not incorporate uncertainty from reconstructing the FES, selecting a model or changing the clustering seed. This scope is recorded with the output.

## 3.12 Software environment and reproducibility

The modular implementation uses NumPy, SciPy, pandas, Matplotlib and scikit-learn; the key numerical and machine-learning libraries have established primary descriptions [27–29]. A locked environment records installed versions separately from published simulation versions. Snakemake expresses stage dependencies, while CLI commands provide the same functionality without notebooks [4, 30].

Each run begins with a manifest and resolved parameters. A supervisor records the worker's status and samples process-tree memory every 0.1 s, enforcing configured runtime and memory limits. Successful, failed, and blocked runs have distinct final statuses. Figures may be rebuilt only from a successful manifest after its output hashes are checked. The resulting record links each figure to the calculation and data that produced it.

# 4 Reproduction and scientific code audit

## 4.1 Original implementation

The supplied implementation is a monolithic script whose top-level execution reads inputs, writes segmented replica files, creates figures, performs clustering, and computes summary tables. Static parsing identifies an un-commented Italian sentence at line 437, causing a syntax error before the analysis can run. The original is preserved exactly, including this defect, so that the audit can be checked against the supplied file.

The audit follows each source operation from input to reported quantity. Static inspection establishes what the script attempts to do, but no original execution manifest or complete numerical output set was located. Where source behavior differs from the paper, the method map records both descriptions and the correction used in the modular implementation.

## 4.2 Identified methodological issues

The original parser silently takes the first three reset segments and skips certain malformed lines. The new parser checks segment count and row validity. Original group generation depends on a fixed list of 17 seeds, which would be exhausted under smaller group sizes. The replacement uses local RNG streams and retains a complete identity map. This is a reproducibility improvement but changes exact sampled subsets, so bitwise equivalence to the legacy script is not claimed.

Residual handling changes which observations define a family. In the supplied code, rejected frames receive a generic label and every label enters consensus. Frames that failed the acceptance criteria can consequently influence the global centers. The modular version admits only accepted memberships to discovery, then assigns the residual frames in STEP 3. This preserves their contribution to the final population without treating them as an accepted local cluster.

The FES reader ignores completeness metadata, permits missing grid values with a warning, and the weight function uses nearest-grid lookup with boundary selection. These behaviors can generate plausible but unsupported weights. The replacement rejects incomplete grids and performs bounded bilinear interpolation. The original partial download ended inside a numerical row and was rejected. The complete replacement received on 7 October passed grid validation; it did not retroactively make the partial file usable.

The original weighted-analysis section applies weights to relevance but calculates arithmetic means and standard deviations for the structural descriptors. Those summaries describe a different distribution from the weighted one specified in the paper's figure captions. The revised routines use consistent weights for relevance and conditional descriptor moments, and normalize replicas separately before pooling. Distribution width, replica SEM and bootstrap intervals remain separate quantities.

**Scientific audit note**

Correcting a code path does not establish the validity of its input data or the convergence of the original simulation. The complete grid now passes validation, but its energy-unit and construction provenance still require clarification before empirical reweighting.

## 4.3 Scientific V2 corrections: evidence unavailable

The requested intermediate file `CLUSTERING40_scientific_v2.py` and its changelog were not located. The method map therefore marks its implementation column as unavailable throughout. None of the corrections in this report is attributed retrospectively to that version. If it is later supplied, it should be hash-verified, preserved without modification, and compared with both the original and the modular implementation.

## 4.4 Verification and numerical tests

The current suite passed 36 tests, including strict parsing, interval boundaries, normalization, analytic bilinear interpolation, incomplete and duplicate grids, finite/bounds checks, weight sign, free-energy-offset invariance, equal replica masses, weighted moments, residual handling, final assignment, label-permutation agreement, and bootstrap separation of replicas. A seeded autoregressive fixture checks that the correlation estimator behaves plausibly for a known stationary process.

A complete synthetic pipeline fixture exercises cluster discovery, consensus, all-frame assignment, final-bias weights, population tables, conditional bootstrap intervals and PDF generation. Its generated descriptors and analytic surface are confined to testing and are not treated as Aβ40 observations. An additional repeatability test fits the GMM extension to explicitly synthetic separated observations. Its label-permutation and missing-family tests verify comparison behavior. An archive-provenance test also checks that a source release cannot inherit the Git commit of an unrelated enclosing repository. These tests establish numerical behavior, not the validity of a biological family.

## 4.5 Remaining methodological ambiguities

The main unresolved choices are the exact STEP-2 descriptor scaling, the construction of the replica-averaged FES, and the detailed convention for published cluster widths and uncertainties. Density ties and spline-based elbow selection are also more explicit in the supplied code than in the paper. The repository records these as source-informed choices rather than presenting them as fully specified published theory.

These ambiguities have consequences for implementation and comparison. Raw Wasserstein scaling remains the reproduction setting; normalized scaling has its own extension configuration. FES construction remains unresolved even though the numerical grid is complete. Before a future discrepancy in populations is interpreted biologically, differences in these analysis choices will need to be assessed.

# 5 RobustCluster-Aβ: robustness extension

## 5.1 Temporal correlation and effective sampling

For each replica and baseline descriptor, the extension computes a demeaned autocorrelation sequence using an FFT-based convolution. Covariances use the common $1/N$ normalization. Adjacent positive-lag terms are paired; the sum is truncated at the first nonpositive pair and monotonized by cumulative minima. The resulting diagnostic is

$$g = \max\left(1, 1 + 2 \sum_{k \in \mathscr{K}} \hat{\rho}(k)\right), \qquad N_{\mathrm{eff,corr}} = N/g, \qquad \tau_{\mathrm{int}} = g\Delta t/2. \tag{5.1}$$

In eq. (5.1), $\mathscr{K}$ denotes the retained lag contributions after the positive-sequence processing; the implementation saves the truncation lag. The convention includes the zero-lag half term in $\tau_{\mathrm{int}}$. Different conventions in the literature differ by that term, so reporting only a number called "correlation time" would be ambiguous.

Three checks qualify the estimate: whether a truncation was observed, whether $N \geq 50g$, and whether the absolute shift between first- and second-half means is at most 0.25 of the full-series standard deviation. Failure of any check flags the result. These screening heuristics indicate when the available series provides weak support for the estimate; they do not constitute a formal stationarity test. Sampling-quality analyses likewise require the data and assumptions to support the chosen uncertainty estimator [11, 12].

Weight concentration is assessed separately by

$$N_{\mathrm{eff,Kish}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{5.2}$$

Equation 5.2 describes how unequal weights reduce the effective number of contributing observations, assuming no additional dependence correction. It is not multiplied by, or automatically divided by, the correlation ESS. A single combined ESS would require an estimator-specific treatment of dependence in the weighted observable. No real Kish ESS is reported before validated statistical weights exist.

## 5.2 Cluster stability across seeds

Random grouping affects which local clusters are discovered, so the planned DPC seed experiment repeats discovery, consensus cutting and final assignment together. Relabeling a fixed partition would leave that source of variation untested. The smoke profile uses two seeds, the separate pilot uses five, and the held full profile defaults to 20, with support for a 20–50-seed study.

Partitions are compared on identical source-frame identities. ARI assesses agreement independent of label names [31]. A maximum-overlap assignment then matches families across runs, after which Jaccard overlap, population variation and centroid movement can be summarized. Unmatched families are retained explicitly, including the appearance or disappearance of a family. A small average variation among only the matched families would otherwise conceal instability in cluster number.

Centroid comparisons across representations use the original four descriptors, rescaled by the common baseline ranges. Thus a PCA-space centroid is not compared directly with a raw-space centroid. Population variation across seeds describes algorithmic sensitivity under the same input ensemble; it is not interchangeable with sampling uncertainty between independent replicas.

## 5.3 Feature redundancy and PCA

The completed lightweight diagnostics calculate the pooled Pearson correlation matrix and PCA after standardizing each baseline feature to zero mean and unit variance. The explained-variance spectrum measures the geometry of the observed descriptor cloud. It does not measure thermodynamic relevance or identify a slow physical mode.

The alternative clustering configurations retain all principal components, optionally followed by whitening. With all components retained, PCA is an orthogonal rotation of standardized variables and preserves their Euclidean distances. A difference from the min–max baseline would therefore reflect standardization, whitening, numerical effects or a downstream choice, rather than information loss through dimension reduction. The distinction is necessary when interpreting any eventual change in clustering.

## 5.4 Independent GMM feature-ablation experiment

The completed original experiment compares the four baseline descriptors with $n_\alpha$, $n_\beta$ and $R_g$ after removing SASA. It uses Gaussian mixtures in scalar descriptor space, a different model from both the published DPC pipeline and coordinate-based ShapeGMM. For a standardized observation $\mathbf{x}$,

$$p(\mathbf{x}) = \sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \Sigma_j), \qquad \sum_j \pi_j = 1. \tag{5.3}$$

Equation 5.3 approximates the sampled descriptor distribution. Its mixture coefficients are not equilibrium statistical weights. The implementation uses scikit-learn [29]; full covariance matrices allow dependencies between features within a component.

The experiment uses one fixed, replica-balanced fitting sample: 1,000 frames from each replica, selected with sampling seed 20261007. Feature means and standard deviations are fitted on those 3,000 rows and reused for the shared features in both representations. At each $K = 2, \ldots, 6$, five seeds (42–46) initialize separate models, each with a 500-iteration limit, tolerance $10^{-3}$ and covariance regularization $10^{-6}$. Using one initialization per fit exposes variation between local solutions instead of selecting the best of several starts. The protocol was committed before execution.

All 300,003 observations receive a label from every fitted model, including the observations used for fitting. ARI then compares the four- and three-feature assignments at each seed and component count; the ten seed pairs are compared within each representation. Maximum-overlap matching adds an interpretable correspondence between labels, with Jaccard overlap, unmatched families and centroid shifts measured in the shared standardized three-feature space. Replica-specific raw label fractions are also retained. This is a descriptive comparison on the same data, not a held-out predictive test; the ranges over runs are not confidence intervals.

This experiment holds the fit subset fixed to isolate representation and initialization effects. It does not quantify sample-selection uncertainty or remove temporal dependence. BIC is stored for comparisons within a feature space only; it is not compared across dimensions or used to declare a physical number of states. The planned DPC ablations remain separate experiments with their own provenance requirements.

## 5.5 Wasserstein scaling sensitivity

The raw consensus distance and the normalized alternative use the same marginal-distance construction, changing only the descriptor scaling used in that stage. This isolates a specific implementation ambiguity. Because raw SASA and radius-of-gyration values have different units and numerical ranges, an unscaled aggregate norm can be influenced disproportionately by the larger-range variable.

The normalized comparison would quantify the consequence of the scaling convention, while leaving the question of the original authors' intended implementation open. Joint distances, sliced Wasserstein, energy distance and MMD could test other aspects of distributional similarity in later studies. None is substituted for the current baseline.

## 5.6 Reweighting sensitivity

The optional comparison holds the fitted partition fixed and contrasts the final-bias estimator with weights derived from verified `metad.rbias`. It compares reweighted distributions of all four baseline

descriptors and family relevance. This separation attributes any differences to weighting rather than to simultaneous changes in the partition.

The descriptor file contains a column named `metad.rbias`, but neither the original PLUMED configuration nor its energy-unit evidence is available. The optional comparison therefore remains blocked at input validation. Its implementation follows the distinction between estimators described in the primary literature and official PLUMED reference [16–18].

## 5.7 Alternative density-based clustering

DPA is a high-value future benchmark because it provides a different treatment of density estimation and significant peaks [21]. Its evaluation should compare family number, location, population and correspondence using the same observation identities and validated weights. No DPA fit has been performed in this project.

Weighted ShapeGMM is relevant to biased structural ensembles [22]. However, applying it as a coordinate-based comparison requires atomic structures, a consistent atom selection and alignment convention, and valid frame mapping. Scalar descriptor rows cannot be expanded into those structures. The repository therefore leaves this option explicitly deferred.

## 5.8 Optional external validation

The experiment-constrained Aβ40 ensemble literature offers an opportunity to compare global compaction and secondary-structure behavior with an independent source [7]. Before numerical comparison, sequence, temperature, weighting, atom selections and observable definitions must be checked. A mass-weighted $R_g$ and a Cα-only $R_g$, for example, need not have directly comparable distributions.

No external ensemble has been processed in this project. The feasibility assessment identifies the relevant study and compatibility requirements; testing agreement with NMR/SAXS-constrained descriptors remains future work.

# 6 Results

## 6.1 Available evidence and execution status

Input validation established three segments of 100 001 rows, each at the source-labelled 20 ps spacing. The table passed the schema, finite-value and interval checks used by the analysis. These checks support the scalar-descriptor calculations below; simulation convergence and upstream descriptor definitions remain outside what can be established from the table alone.

Run `20260921T011959-583b4421` produced the descriptor diagnostics in 2.28 s, with sampled peak worker-process memory of 418.2 MiB. These are measurements of a lightweight local calculation, not projections of full DPC cost. The status summary in table 1 separates those diagnostics, the subsequent complete-grid FES check and the GMM pilot from the blocked reproduction attempt.

| Analysis | Evidence status |
|---|---|
| Input validation | Completed on the supplied Aβ40 descriptor table |
| Correlation matrix / PCA | Completed; unweighted biased-sample diagnostics |
| Autocorrelations / provisional ESS | Completed; every estimate flagged as unreliable |
| FES grid and interpolation | Completed; units/construction remain unverified |
| Weighted paper reproduction | Blocked: unit/construction provenance |
| GMM seed / SASA-ablation pilot | Completed; biased-sample geometric comparison |
| DPC seed / feature / scaling studies | Not completed; separate reproduction prerequisites |
| Time-dependent weight comparison | Blocked: FES and rbias unit/provenance requirements |
| Molecular representatives / ShapeGMM | Blocked: coordinate trajectory and mapping absent |
| External validation / Leonardo jobs | Not performed |

**Table 1:** Scientific evidence available at release. Tests on synthetic fixtures are excluded from the empirical analysis statuses.

## 6.2 Descriptor distributions and redundancy

The pooled Pearson correlation is 0.848 for Rg and SASA and -0.456 for the two secondary-structure descriptors. The strong Rg–SASA association is the starting observation for the ablation experiment: both variables describe aspects of molecular dimensions, yet correlation alone leaves their role in assigning individual frames unresolved.

The first two standardized principal components explain 51.8% and 36.3% of total variance, respectively, for a cumulative 88.1% (fig. 3). Much of the descriptor variance is therefore concentrated in two directions. Whether lower-variance distinctions matter for a fitted partition is assessed through the clustering comparison, rather than decided from the spectrum.

Figure 4 compares replicas using common bin edges for each descriptor. Each histogram is normalized by its own frame count and bin width, so differences in shape reflect the supplied samples. The curves have no equilibrium correction and cannot by themselves establish differences in equilibrium populations.

## 6.3 Temporal correlation and provisional ESS

The correlation diagnostics produce provisional ESS values ranging from 3.2 to 28.7 across the 12 replica–descriptor series, compared with 100 001 raw frames in each series. All 12 estimates fail at least one configured reliability heuristic. They should therefore be read as warning indicators of strong dependence and possible nonstationarity, not as precise counts of independent equilibrium conformations.

Table 2 gives the estimator outputs alongside the shift in means between the two halves of each series. Several descriptors change substantially over that split. Together with the reliability flags, this makes

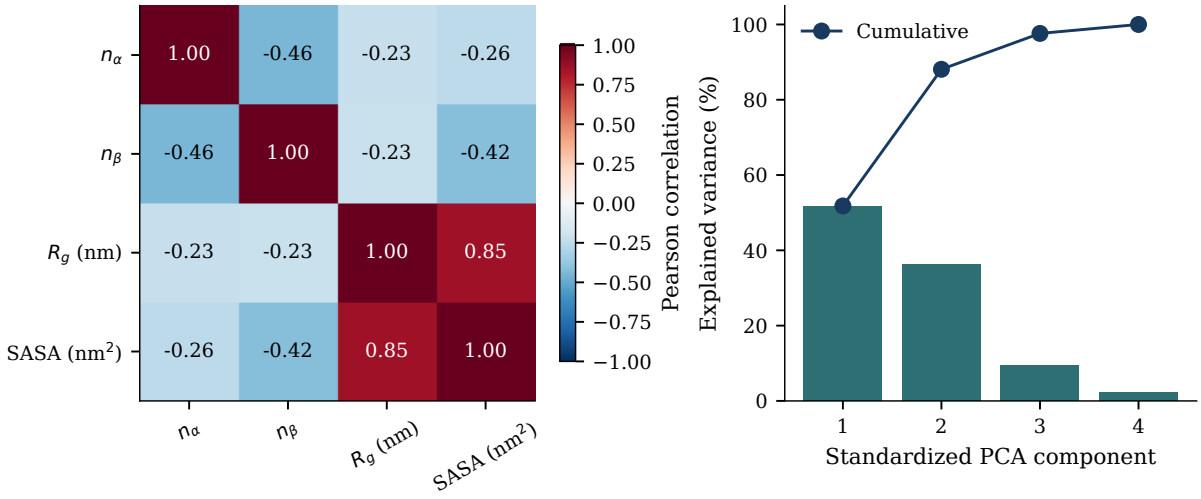



**Figure 3:** Computed diagnostics for the pooled, unweighted Aβ40 descriptor sample. Left: Pearson correlations of the four baseline descriptors. Right: explained variance of PCA after per-feature standardization, with cumulative variance shown by the line. These are geometric summaries of the biased sample and do not represent reweighted equilibrium probabilities.

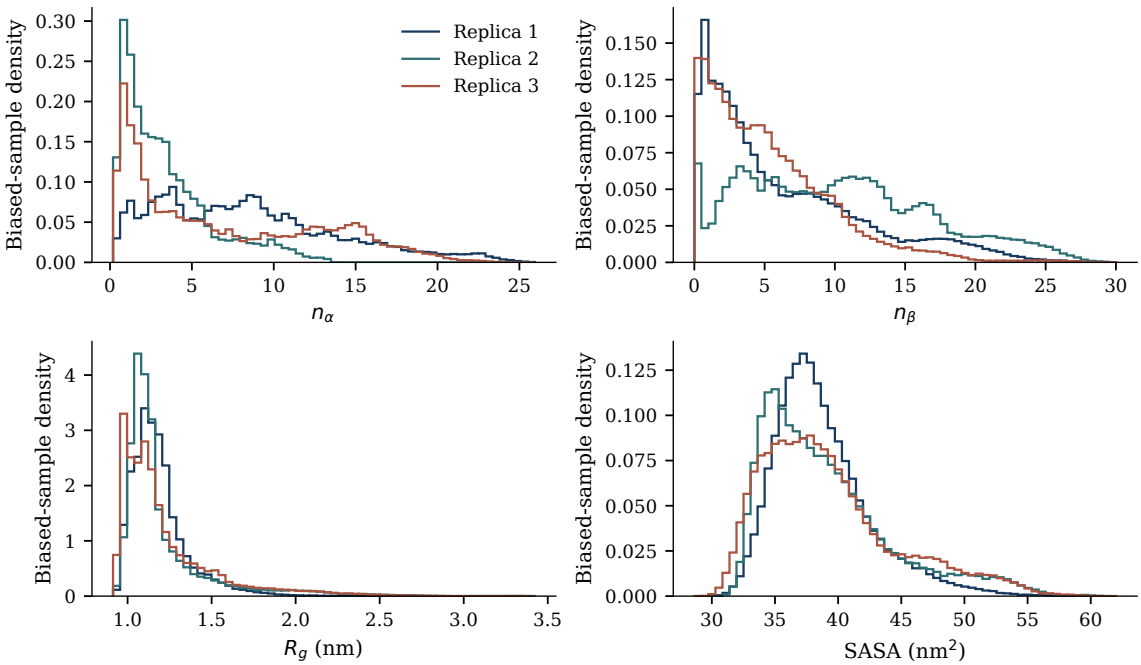



**Figure 4:** Actual per-replica descriptor histograms over the supplied interval. Curves use common bin edges for each descriptor and independent density normalization. No final-bias or time-dependent statistical weights have been applied. Units follow the original script's labels and remain a recorded upstream-provenance assumption.

an independent-frame uncertainty calculation poorly supported. Assessing the published convergence analysis would require the full bias history and its original diagnostics, which are unavailable here.

| Replica | Descriptor | $g$ | $\tau_{\mathrm{int}}$ (ns) | $N_{\mathrm{eff,corr}}$ | Half shift / SD |
|---|---|---|---|---|---|
| 1 | $n_\alpha$ | 22067 | 220.7 | 4.5 | 1.28 |
| 1 | $n_\beta$ | 11079 | 110.8 | 9.0 | 0.83 |
| 1 | $R_g$ | 3679 | 36.8 | 27.2 | 0.27 |
| 1 | SASA | 3490 | 34.9 | 28.7 | 0.36 |
| 2 | $n_\alpha$ | 6259 | 62.6 | 16.0 | 0.37 |
| 2 | $n_\beta$ | 10843 | 108.4 | 9.2 | 0.41 |
| 2 | $R_g$ | 3951 | 39.5 | 25.3 | 0.33 |
| 2 | SASA | 5325 | 53.2 | 18.8 | 0.17 |
| 3 | $n_\alpha$ | 31742 | 317.4 | 3.2 | 1.61 |
| 3 | $n_\beta$ | 5880 | 58.8 | 17.0 | 0.71 |
| 3 | $R_g$ | 6259 | 62.6 | 16.0 | 0.88 |
| 3 | SASA | 9958 | 99.6 | 10.0 | 0.87 |

**Table 2:** Computed correlation diagnostics. All entries are flagged as unreliable under the stated heuristics. The time scale is measured along the biased trajectory, using the source-labelled sampling interval. $g$ is statistical inefficiency; the final column is the absolute difference of half-window means divided by the full-series standard deviation. No physical relaxation-time interpretation is made.

## 6.4 Complete FES validation

The recovered file contains exactly 2,801 by 1,301 grid points: 3,644,101 rows with five finite columns, no duplicate coordinates and complete Cartesian coverage. Its nonperiodic axes span 0–28 and 0–13 at spacing 0.01. All 300,003 converted descriptor queries lie within those bounds, and bounded bilinear interpolation returns finite values for every query. The grid check is an input-integrity result; the energy column has no declared unit and is not assigned one solely because the source script assumes kJ/mol.

## 6.5 Representation and initialization sensitivity

All 50 models in the GMM pilot met their configured numerical convergence criterion. The run took 7.01 s, with a sampled process-tree peak of 660.2 MiB. Despite this common stopping status, the fitted assignments differ across component counts and initializations (fig. 6). Convergence of the fitting algorithm and stability of the partition are therefore separate observations. At three components,

| Components | Median ablation ARI | Observed range |
|---|---|---|
| 2 | 0.690 | 0.188–0.868 |
| 3 | 0.864 | 0.862–0.868 |
| 4 | 0.607 | 0.548–0.622 |
| 5 | 0.531 | 0.382–0.696 |
| 6 | 0.457 | 0.294–0.461 |

**Table 3:** Observed representation agreement across five paired initializations. Component count was varied by design; the table does not identify an optimal biological family count.

cross-representation ARI lies between 0.862 and 0.868, with approximately 95.3% of frames retaining corresponding labels after maximum-overlap matching (median across seeds). Within-space seed agreement is also high: 0.994–1.000 with four descriptors and 0.988–1.000 without SASA. At two components, however, within-space ARI ranges down to 0.011 and 0.067, respectively. A single initialization would conceal this variability. The full matrices are retained in fig. 7.

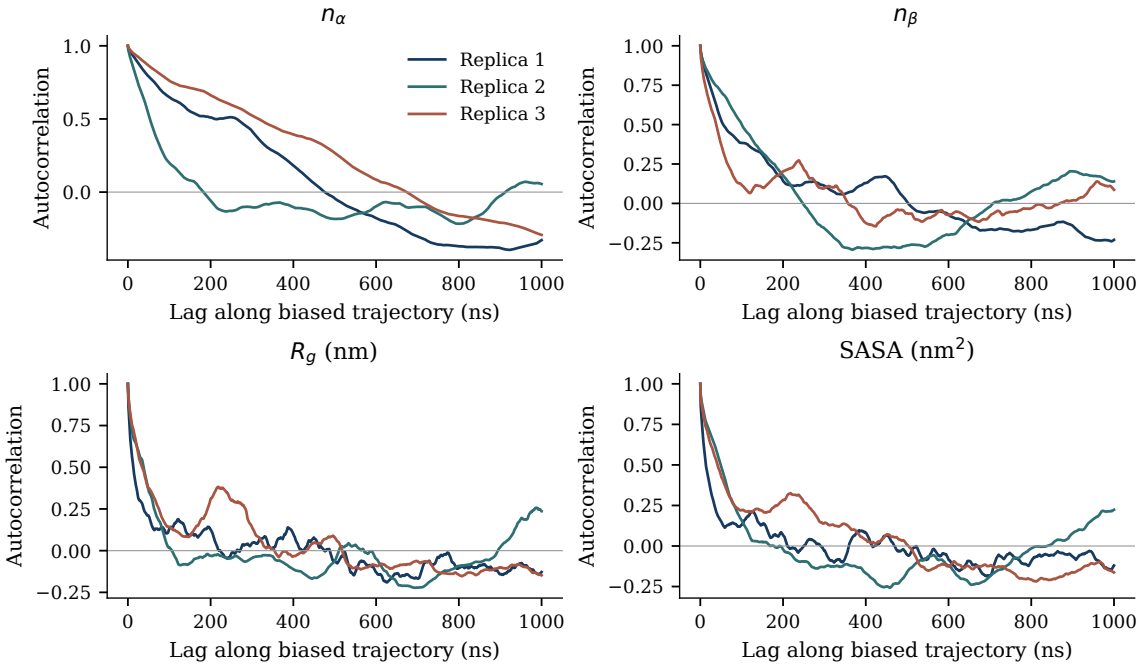



**Figure 5:** Autocorrelation functions computed separately for each replica and descriptor. The plotted curves cover the first half of the available lag window with a reduced plotting density; the estimator operates on the full, un-decimated ACF sequence. Long persistence and changing behavior across replicas reinforce the need for cautious uncertainty analysis. Lag is biased trajectory time, not recovered physical time.

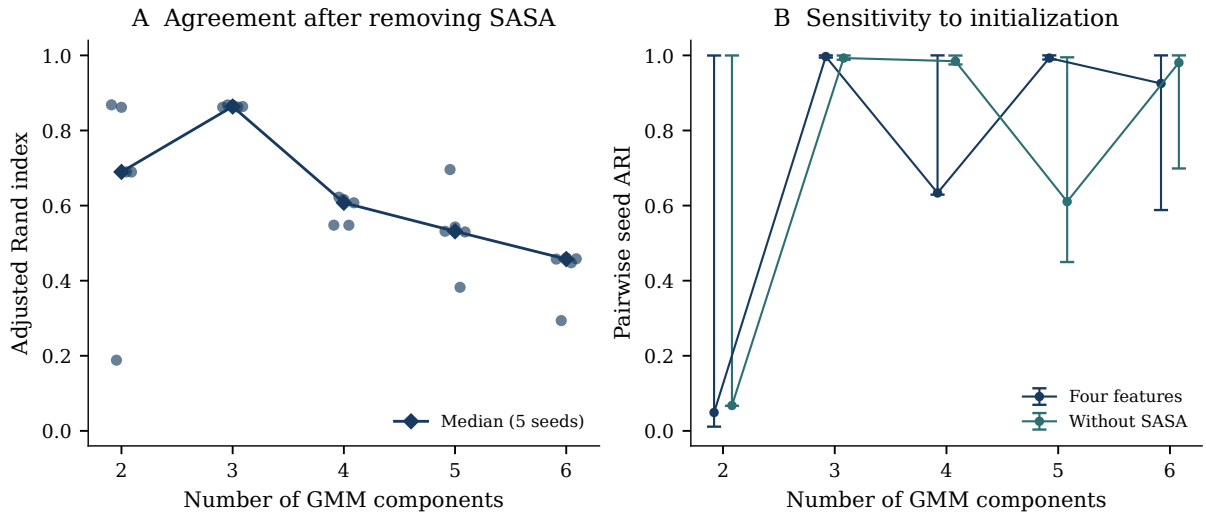



**Figure 6:** Independent GMM sensitivity experiment on the biased descriptor sample. Left: four-feature versus SASA-ablated assignments at each of five seeds; diamonds mark medians. Right: within-space ARI over ten seed pairs; bars span observed minima and maxima, not confidence intervals. The fit subset is fixed across all models and evaluation includes fit rows.

The most consistent preservation of assignments in this pilot occurs at three components. At the other tested counts, the effect of removing SASA is larger or more dependent on initialization. This confines the conclusion to a specified model resolution and fit design: it does not support dropping SASA from the published baseline. The comparison concerns assignments in the biased sample; no equilibrium weighting was applied.

## 6.6 Analyses remaining unavailable

Equilibrium cluster relevance, weighted family characterization and completed paper-DPC robustness remain unavailable. No molecular snapshots are created because the descriptor table has no atomic coordinates. The completed GMM experiment is an original descriptor-space extension, not a replacement for those stages.

# 7 Discussion

## 7.1 What the completed experiments support

The ablation experiment resolves a question left open by the correlation matrix. Rg and SASA are strongly associated across the pooled sample, but that association does not determine the fitted partition after SASA is removed. Three-component mixtures give high cross-representation agreement and high agreement between initializations. Other component counts are less consistent. The effect of feature removal is therefore conditional on the model being fitted, rather than a property that can be inferred from one correlation coefficient.

The fit design limits how far this result can be generalized. All models use the same selected observations, so the experiment isolates representation and initialization while leaving selection of the fitting sample untested. Randomly choosing those frames does not make the underlying trajectory independent. Every replica–descriptor series fails at least one correlation-estimator reliability criterion; the provisional ESS values consequently indicate a sampling concern rather than calibrated ensemble sizes. New fitting subsets and time-block comparisons would be needed to examine sampling sensitivity [12].

## 7.2 Relation to the published study

The repository reconstructs the published analysis design, but has not reproduced its reweighted conformational families or their populations [2]. The completed GMM pilot is an independent descriptor-space extension. It provides a methodological comparison, not validation of published biological conclusions. Neither the diagnostic correlations nor stable Gaussian assignments establish aggregation rates, toxicity or metastable states. Comparison with an independently constrained ensemble would additionally require compatible structures and observable definitions [7].

## 7.3 Limitations and methodological implications

Two levels of model adequacy remain unresolved. First, a converged mixture fit can correspond to a local solution, and stability across seeds only describes the starts that were tested. Second, a reproducible partition of scalar descriptors need not identify distinct physical states. Atomic representatives, contacts and alternative descriptors could help assess the latter question if structural trajectories become available. In the separate consensus baseline, comparing marginal distributions also leaves sensitivity to joint dependence unresolved.

The FES poses a different limitation. Its grid is complete and covers every query, but its energy units, replica combination and bias convergence have not been established. Once weighting becomes possible, concentration of the weights will introduce a further source of variability alongside temporal dependence and initialization. Three replicas provide only a small set of independent replica-level comparisons, which will constrain uncertainty assessment.

The current evidence nevertheless supports specific reproducibility tasks. Saved labels allow the GMM agreement calculations to be checked without refitting, and source tables allow the diagnostic figures to be rebuilt. Reproducing equilibrium family populations requires the missing FES information and a new calculation. This separation makes the available work useful to another researcher without asking that reader to infer unavailable provenance [6, 32].

# 8 Internship outcomes and skills

## 8.1 Learning sequence and practical objectives

The internship linked Data Analysis methods with the interpretation of molecular simulations. Its documented learning programme began with amino-acid properties, peptide bonds, protein structure and intrinsic disorder, then introduced force fields, molecular dynamics and dihedral angles through supplied slides and readings. The final assignment brought these topics together through reproduction of a clustering analysis, investigation of parameter changes and an independent analysis using methods from the degree.

Two distinctions shaped the practical analysis. Consecutive rows describe saved configurations, but they need not supply independent observations. Likewise, the number of frames assigned to a group describes its frequency in the recorded sample, while its equilibrium probability requires valid statistical weights. Both are familiar data-analysis questions whose interpretation changes when the data come from a biased molecular simulation.

## 8.2 Demonstrable analysis and software outputs

The completed outputs include an input and code audit, modular numerical routines, 36 passing tests, real descriptor diagnostics, a complete-grid FES check, and an independent GMM sensitivity experiment. The latter turns a descriptive observation—the Rg–SASA correlation—into a testable question about assignments. Retaining all component counts and initializations allows both stable and sensitive cases to be examined.

Feature scaling provides a small example of how a software choice affects the scientific calculation. The model must use the transformation fitted on its designated sample when it assigns later observations. The following searchable excerpt is taken directly from the implementation.

**01 Fit the geometry once** RobustCluster-Aβ / Python

```python
def fit_scaler(fit_values, values):
    fit_values, values = finite(fit_values), finite(values)
    mean, scale = fit_values.mean(axis=0), fit_values.std(axis=0)
    if np.any(scale <= 0):
        raise InvalidInput("Constant fit feature")
    return (values - mean) / scale, {"mean": mean.tolist(), "scale": scale.tolist()}
```

The function rejects a constant fitting feature and returns the parameters used for the transformation. Its test introduces a large evaluation value and checks that the fitted mean and scale remain unchanged. This makes the geometric convention explicit and checks that prediction cannot silently redefine it.

## 8.3 Companion projects: VMD MCP and GROMACS MCP

Two additional public software outputs are VMD MCP and GROMACS MCP. Both provide interfaces around molecular software through the Model Context Protocol (MCP), which exposes operations to a client using structured inputs and outputs. Their inspected code illustrates how a software interface organizes a calculation; the underlying program and its inputs still determine the scientific result.

VMD MCP constructs Tcl commands for operations such as per-frame radius of gyration, RMSD, SASA and rendering. The gyration function passes the selected atoms and mass-weighting option to VMD, then parses frame-indexed values. GROMACS MCP includes helpers for simulation preparation and analysis and a background-run interface that constructs a command, validates paths and records output. Genuine source screenshots and immutable commit links appear in figs. 8 and 9.

The companion projects operate at a different level from RobustCluster's descriptor pipeline. Their source is evidence of software implementation, not evidence that they generated the supplied Aβ40 trajectories. For example, the inspected VMD gyration function documents an angstrom convention, whereas the descriptor table uses source-labelled nanometres. Combining their outputs would require a checked unit conversion and compatible atom selections. Contributor attribution remains in the respective repository histories.

## 8.4 Reproducibility and computational scope

The project demonstrates software practices through concrete artifacts: preserved inputs, separate configuration files, analytic test cases, recorded failures and figures rebuilt from saved tables. A manifest links a plot to its data and settings. Its status also distinguishes a calculation that could not be completed from one that returned a valid zero. These records make the analysis explainable beyond the code itself.

All newly reported calculations are local descriptor analyses. The project includes a parameterized Leonardo submission template and official documentation references [33], but no Leonardo job records exist. HPC execution, molecular simulation setup and physical kinetic measurements are therefore outside the claimed computational outcomes.

# 9 Conclusions and future work

The supplied Aβ40 descriptors support a bounded computational study: an audit of the analysis code, diagnostic checks of the sampled data and an independent Gaussian-mixture comparison. The audit traced specific problems in execution, residual-cluster handling, FES interpolation, replica pooling and weighted summaries. The modular implementation addresses them while preserving the original files and recording the behavior covered by its numerical tests.

The completed experiment shows that the consequences of removing SASA depend on component count and initialization, despite strong global Rg–SASA correlation. The three-component partition is largely preserved in the tested fits; this cannot be generalized to every model resolution or fitting sample. Correlation diagnostics also indicate that the raw frame count substantially overstates the support for an independent-observation analysis, although their flagged ESS estimates should not be treated as precise effective counts.

The next step toward the published-workflow reproduction is concrete: verify the recovered FES energy units and replica-combination procedure, then perform a bounded smoke run. Its resource measurements can guide larger runs that retain the published scientific parameters. Repeating the analysis across seeds, features and Wasserstein scaling conventions would then test the sensitivity of the resulting families on common frame identities.

Verified bias-offset information would enable the optional time-dependent reweighting comparison. Atomic trajectories and a valid indexing map would enable representatives and coordinate-based benchmarks using tools such as MDAnalysis and ShapeGMM [22, 34]. External experimental-ensemble validation should proceed only after observable definitions and data rights have been checked.

The present contribution is thus a checkable descriptor-space result and the software needed to pursue the remaining reproduction. Its central finding is that feature correlation does not settle a clustering choice. Equilibrium conformational families remain a separate result to be established once the missing provenance is supplied and the corresponding calculations are completed.

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# A Full parameter table

**Table 4:** Reproduction settings and execution profiles. These parameters are implemented but do not imply a completed empirical run.

| Parameter | Value | Evidence / role |
|---|---|---|
| Replicas | 3 | Primary paper; observed table segmentation |
| Analysis interval | 1–3 μs | Published baseline; inclusive source-labelled ps boundaries |
| Temperature | 310 K | Published weighting parameter; no new MD run |
| Bias factor $\gamma$ | 10 | Published weighting parameter |
| Molar gas constant | 0.008314462618 kJ $mol^{-1}$ $K^{-1}$ | Implementation unit convention |
| CV conversions | $\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$ | Published descriptor scaling |
| Baseline features | $n_\alpha$, $n_\beta$, $R_g$, SASA | Pooled min–max normalization |
| Initial $d_c$ | 1.3 | Published baseline |
| Cutoff multiplier | 0.9 | Published 10% reduction |
| Center-count scan | 10 down to 2 | Published baseline |
| Minimum cluster size | 300 frames | Published acceptance criterion |
| Minimum mean silhouette | 0.3 | Published acceptance criterion |
| Failure stopping count | 5 | Consecutive iterations with no accepted cluster |
| Distance storage | float32 | Source-informed; blockwise construction |
| Consensus | Average linkage | Marginal Wasserstein norm |
| Elbow smoothing | $s = 5$ | Supplied source heuristic |
| Elbow search range | First 90% of merges | Supplied source heuristic |
| Final assignment | One nearest-centroid pass | No iterative refitting |
| Master seed | 42 | Default project configuration; recorded per run |
| Smoke / medium groups | 3,000 / 10,000 | Reduced execution profiles |
| Smoke / medium discovery groups | 2 / 3 | Not full-production evidence |
| Full group size | About 30,000 | All groups; full execution disabled |
| Seed counts | 2 smoke; 5 pilot/medium; 20 full default | Full robustness configurable; not performed |
| Bootstrap draws | 500 | Conditional moving-block extension; not performed on real weights |
| Resource policy | Bounded sequential local workers | Full execution held for review |

## Completed descriptor-only GMM pilot

The following settings belong to the successful independent experiment, not to the published DPC baseline. The fit sample and initializations were recorded before execution.

**Table 5:** Settings of the completed GMM experiment, as recorded in its run manifest.

| Parameter | Value | Evidence / role |
|---|---|---|
| Fit sample | 1,000 per replica | 3,000 rows; shared across models |
| Sampling seed | 20261007 | Local NumPy generator |
| Feature spaces | Four descriptors; omit SASA | Same scaling for shared features |
| Scaling | Fit-sample mean and SD | No evaluation-data refit |
| Components | 2, 3, 4, 5, 6 | All counts retained in comparison |
| Initialization seeds | 42, 43, 44, 45, 46 | Paired across representations |
| Initializations per fit | 1 | 50 total fitted models |
| Covariance | Full | Regularization $10^{-6}$ |
| Convergence settings | 500 iterations; tolerance $10^{-3}$ | All fits converged numerically |
| Evaluation | 300,003 rows | Includes fit rows; descriptive comparison |
| Resource limits | 600 s; 4 GiB | Sequential local worker |

# B Mathematical derivations and estimator checks

## Why the sign of the final-bias correction is negative in FES form

Under a static bias $V$, the observed configuration density is proportional to $\exp[-\beta_T(U+V)]$. Multiplying by $\exp(\beta_T V)$ recovers relative canonical density. The WT relation $V \simeq -(1-1/\gamma)F + C$ therefore gives the negative free-energy sign in eq. (2.2). Applying $\exp(-\beta_T V)$ would reverse the correction. This derivation assumes that using a final bias is appropriate for the analyzed interval; it is not a proof that the supplied trajectories satisfy that assumption.

## Equal-replica mixture and cluster moments

Let $\widetilde{w}_{ri}$ sum to one within replica $r$. The pooled expectation of $A$ is $(1/3)\sum_r \sum_i \widetilde{w}_{ri} A_{ri}$. Cluster relevance uses $A$ equal to a membership indicator. For a conditional cluster mean, divide the pooled weighted sum of the descriptor by the pooled membership mass. Computing three conditional means and averaging them equally instead would answer a different question when cluster populations differ across replicas.

## Bilinear interpolation test

For a cell with fractional coordinates $u, v \in [0, 1]$, interpolation combines its four corner values with weights $(1-u)(1-v)$, $u(1-v)$, $(1-u)v$, and $uv$. Their sum is one. Any surface $a + b\alpha + c\beta + d\alpha\beta$ is reproduced exactly within floating-point tolerance. The test suite includes such a mixed-term surface, reordered input rows, boundaries and invalid-grid cases, providing stronger evidence than checking a single grid node.

# C Additional diagnostics and interpretation

The saved tables contain per-replica raw descriptor counts, means, standard deviations, extrema, common-bin histograms, the full four-feature correlation matrix, PCA transformation parameters and explained variance, and the reported correlation-estimator diagnostics. The ACF figure is a plotting subsample, not the series used to estimate $g$. No weights are present in these real-data diagnostic summaries.

The half-window mean-shift diagnostic is descriptive. A large shift can arise from slow fluctuations, nonstationarity, or changing occupation of sampled regions. Its threshold is an explicit project heuristic. A low shift alone would not establish convergence, and no threshold outcome can recover an unbiased time scale from WT-MetaD time. Future evaluation should preserve these distinctions when adding weighted estimates.

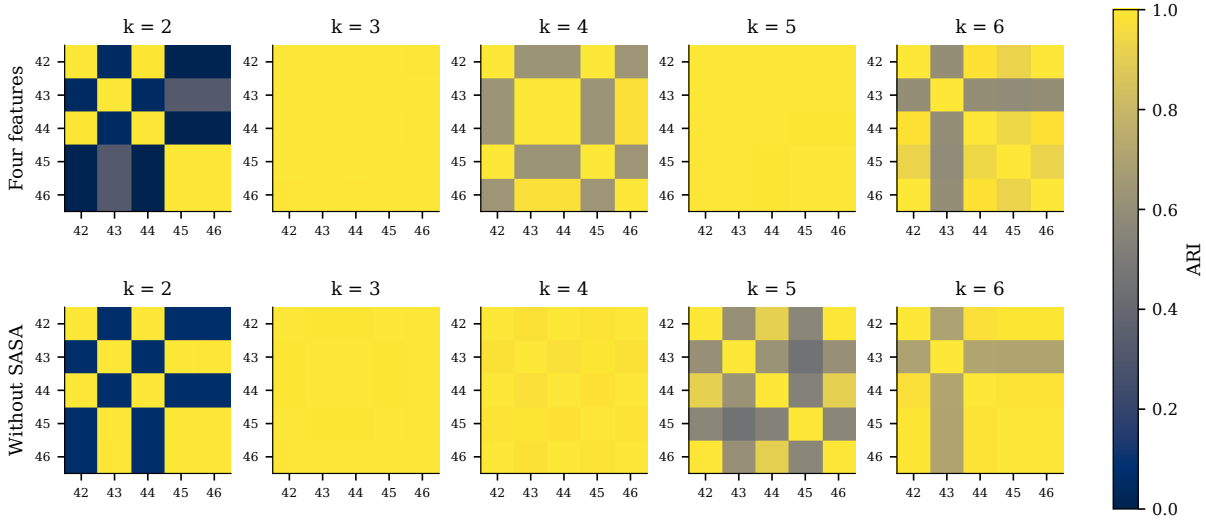



**Figure 7:** All pairwise seed ARIs from the GMM pilot. Columns show each tested component count; rows distinguish the four-feature and SASA-ablated representations. The diagonal is identity. All fits converged, so no entries are omitted. Similar numerical stopping status can coexist with different partitions.

# D Code audit summary

The project inventory records immutable source hashes, table dimensions and missing inputs. The method map links each step to paper evidence, source behavior, the modular implementation and validation tests. The decision log records scientific category and expected effect. The open-questions register prevents uncertain provenance from being converted into hidden defaults.

Scientific V2 is absent, so the audit contains no invented intermediate diff. The supplied original remains excluded from execution and lint collection because its syntax error and top-level writes are part of the preserved evidence. New code is tested independently. Public source packaging excludes the raw table and original script until their redistribution rights are established.

Codex assisted with software refactoring, numerical checks, document preparation and layout. Reported numerical observations come from recorded Python runs; the illustrated cover and two conceptual editorial artworks were generated with the built-in image-generation tool, and the source-code screenshots are actual browser captures. This assistance is separate from the published methodology and the original supplied materials. The generation tool does not expose a verifiable model-version identifier; no specific Image 2.5 execution claim is made. Exact prompts and image hashes are retained in `docs/ARTWORK_PROMPTS.json` and `report/artwork_provenance.json`. The supplied inspiration image guided the print texture and restrained palette, not the scientific content. The cover typography is part of the generated image; its institutional and personal details were visually checked. The University of Messina logo is a separate image supplied by the student and placed without generative reconstruction.

# E Reproducibility instructions

The project source, tests, configurations and this report are available at RobustCluster-Aβ on GitHub. The Python package and LaTeX sources can be inspected separately or downloaded together using GitHub's *Code / Download ZIP* option. The repository guide distinguishes code verification and report regeneration from analyses requiring restricted inputs.

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-lock.txt
python -m pip install --no-deps -e .
pytest -q
ruff check src tests scripts
robustcluster validate --config config/paper_reproduction.yaml
robustcluster diagnostics --config config/robustness.yaml
```

The raw file must first be obtained and checked against its documented SHA256. The standalone commands and Snakemake workflow use the same package functions. The report imports successful diagnostics, FES-audit, GMM and figure manifests with matching hashes. Its generated editorial artwork and real browser screenshots have separate provenance records. LuaLaTeX, Biber and latexmk build the supplied report sources; current paths and tool versions are recorded in the build manifest.

For later production, establish the recovered FES energy units and construction provenance, run smoke validation, review resource measurements, and create a reviewed full-run configuration. HPC templates require an actual allocation and explicit settings. Do not copy account or partition values from unrelated examples, and preserve job evidence before attributing calculations to Leonardo [33].

# F Software and environment table

| Component | Version in diagnostic manifest |
|---|---|
| Python | 3.12.6 |
| numpy | 2.5.3 |
| scipy | 1.18.1 |
| scikit-learn | 1.9.1 |
| pandas | 2.3.3 |
| matplotlib | 3.11.2 |
| robustcluster-abeta | 0.1.0 |

**Table 6:** Actual software versions for the reported local descriptor diagnostics; the GMM pilot manifest records the same versions. They do not describe the original authors' simulation environment. The complete dependency lock and report-tool versions are separate reproducibility artifacts.

The checked bibliography contains journal articles and official documentation records; its matrix links each citation to its question, method, relevance, and report section. Online-first dates and issue years are distinguished. References whose publisher verification remained unresolved are documented outside the BibTeX file rather than filled from memory.

# G Companion source-code evidence

The following images are genuine browser captures from public, commit-specific GitHub source pages on 7 October 2026. They document implementation, not simulation execution. The full images are preserved unchanged; short searchable listings in the main text are a separate presentation format.

**02 Measure each saved frame** — VMD MCP / source lines 250–266

```
250      weight = "weight mass" if mass_weighted else ""
251      tcl = f"""
252  {_load_block(structure, trajectory)}
253  set sel [atomselect $mol "{_tcl_quote(selection)}"]
254  set n [molinfo $mol get numframes]
255  for {{set i 0}} {{$i < $n}} {{incr i}} {{
256    $sel frame $i
257    puts "{MARKER} frame=$i rgyr=[measure rgyr $sel {weight}]"
258  }}
259  quit
260  """
261      res = _run_tcl(tcl)
262      series = [
263          {"frame": int(m["frame"]), "rgyr": float(m["rgyr"])}
264          for m in res.get("markers", [])
265          if "rgyr" in m
266      ]
```

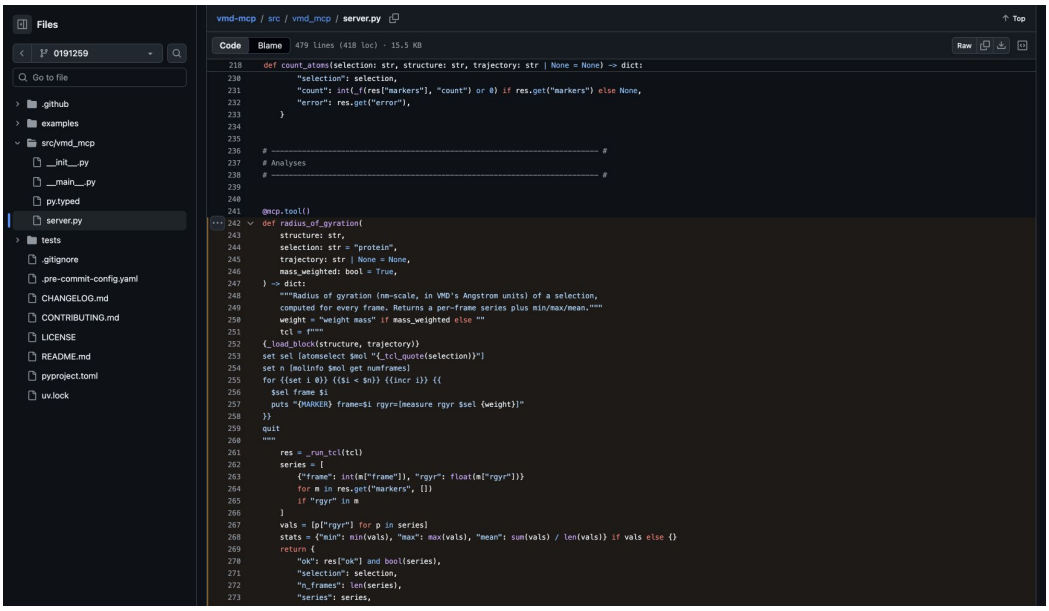



**Figure 8:** VMD MCP source at commit 0191259: the gyration wrapper constructs a Tcl frame loop and parses marked values. The screenshot shows the repository tree, source path and real line numbers. The function's documented unit convention must be checked before comparison with other tools. Open the inspected source.

**03 Validate before starting a job** GROMACS MCP / source lines 1452–1467

```
1452        wd, err = _workdir_or_error(workdir)
1453        if err:
1454            return err
1455        assert wd is not None
1456        if ntomp is not None and ntomp <= 0:
1457            return _error("ntomp must be positive when provided")
1458        if nsteps is not None and nsteps < 0:
1459            return _error("nsteps must be non-negative when provided")
1460        safe_tpr = _safe_file_arg(wd, tpr, "tpr", must_exist=True)
1461        safe_deffnm = _safe_deffnm(deffnm)
1462        args = [GMX_BIN, "mdrun", "-v", "-s", safe_tpr, "-deffnm", safe_deffnm]
1463        if ntomp is not None:
1464            args += ["-ntomp", str(ntomp)]
1465        if nsteps is not None:
1466            args += ["-nsteps", str(nsteps)]
1467        args += _validate_arg_tokens(extra_args, "extra_args")
```

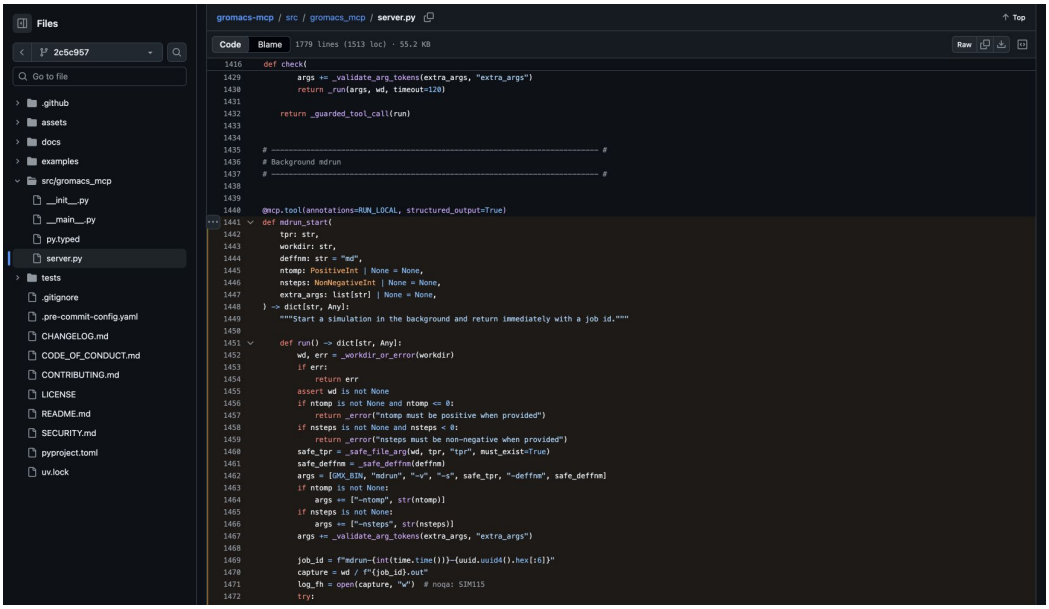



**Figure 9:** GROMACS MCP source at commit `2c5c957`: the background-run wrapper validates inputs, constructs the `mdrun` command and prepares an output log. This source view does not establish that an Aβ40 simulation was launched through the tool. Open the inspected source.